

Product Data Exploration using Microsoft Excel

- Hishaam J

Formulas used in this dataset:

1. SUM = SUM(D2:D35)
2. COUNTA =COUNTA(B2:B35)
3. AVERAGE =AVERAGE(D2:D35)
4. MINIMUM =MIN(D2:D35)
5. MAXIMUM =MAX(D2:D35)
6. PRICE RANGE =IF(D2>500,"High Price"."Standard Price")
7. DAY =LEFT(A2,2)
8. COUNTRY CODE =RIGHT(A2,2)
9. MONTH =MID(A2,4,3)
- 10.SUMIF =SUMIF(range, criteria, [sum_range])
- 11.COUNTIF =COUNTIF(range, criteria)

The screenshot shows an Excel spreadsheet with two columns of formulas. Column 1 (M) contains formulas for Sum, Count, and Average. Column 2 (R) contains formulas for Minimum and Maximum values. The formulas are: SUM(D2:D35), MIN(D2:D35), COUNTA(B2:B35), MAX(D2:D35), and AVERAGE(D2:D35). The results are displayed below each formula.

	M	R
3		
4	1. Sum, Count and Average	2. Minimum and Maximum values
5		
6	Calculate the total price of all products	Identify the minimum price among all products
7		
8	FORMULA =SUM(D2:D35)	FORMULA =MIN(D2:D35)
9		
10	Total Price \$ 10,100	Minimum Price 30
11		
12		
13		
14	Determine how many products exist	Identify the maximum price among all products
15		
16		
17	FORMULA =COUNTA(B2:B35)	FORMULA =MAX(D2:D35)
18		
19	Total Count 34	Maximum Price 1000
20		
21		
22	Calculate Average price of the products	
23		
24		
25	FORMULA =AVERAGE(D2:D35)	
26		
27		
28	Average 297.058824	
29		
30		
31		

4. Conditional Functions - SUMIF and COUNTIF

Use the SUMIF function to calculate the total price of products in the Electronics category.

FORMULA: =SUMIF(range, criteria, [sum_range])

Total Price 8050

Use the COUNTIF function to determine the number of products with a price less than \$100.

FORMULA: =COUNTIF(range, criteria)

Number of Products 11

G2: =IF(D2>500,"High Price","Standard Price")

	A	B	C	D	E	F	G	H	I	J	K	L	M
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR			
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR			
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY			
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN			
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL			
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG			
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard Price	05	UK	SEP			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT			
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV			
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC			
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB			
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR			
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG			
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG			
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN			
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR			
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG			
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY			
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN			
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL			
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG			
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP			
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN			

1. Sum, Count and Average

Calculate the total price of products in the Electronics category.

FORMULA: =SUMIF(range, criteria, [sum_range])

Total Price \$

Determine how many products have a price less than \$100.

FORMULA: =COUNTIF(range, criteria)

Total Count

Calculate Average price of products in the Electronics category.

FORMULA: =AVERAGEIF(range, criteria, [average_range])

H2	=LEFT(A2,2)													
	A	B	C	D	E	F	G	H	I	J	K	L	M	
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN				
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB				
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR				1. Sum, Count and Average
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR				
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY				Calculate the total price of all products
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN				
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL				
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG				FORMULA =SUM
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard Price	05	UK	SEP				Total Price \$
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT				
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN				
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV				
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC				Determine how many products are sold in each month
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB				
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR				FORMULA =COUNT
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG				Total Count
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG				
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN				
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR				
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG				
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY				Calculate Average price of products
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN				
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL				FORMULA =AVERAGE
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG				
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP				
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN				

I2	=RIGHT(A2,2)													
	A	B	C	D	E	F	G	H	I	J	K	L	M	
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN				
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB				
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR				1. Sum, Count and Average
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR				
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY				Calculate the total price of all products
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN				
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL				
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG				FORMULA =SUM
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard Price	05	UK	SEP				Total Price \$
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT				
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN				
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV				
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC				Determine how many products are sold in each month
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB				
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR				FORMULA =COUNT
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG				Total Count
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG				
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN				
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR				
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG				
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY				Calculate Average price of products
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN				
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL				FORMULA =AVERAGE
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG				
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP				
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN				

J2

fx

=MID(A2,4,3)

	A	B	C	D	E	F	G	H	I	J	K	L	M
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26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP			
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN			

Dataset

1. Sum, Count and Average

Calculate the total price of all items

FORMULA =SUM

Total Price \$

Determine how many products are in each category

FORMULA =COUNT

Total Count

Calculate Average price of each category

FORMULA =AVERAGE

27°C Mostly clear

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